Question ID: b76a2815

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$$P = \frac{W}{t}$$

The power P produced by a machine is represented by the equation above, where W is the work performed during an amount of time t. Which of the following correctly expresses W in terms of P and t ?

Answer

A. $W = Pt$

B. $W = \frac{P}{t}$

C. $W = \frac{t}{P}$

D. $W = P + t$

Correct Answer: A

Rationale

Choice A is correct. Multiplying both sides of the equation by t yields $P \cdot t = (\frac{W}{t}) \cdot t$, or $Pt = W$, which expresses W in terms of P and t. This is equivalent to W = Pt.

Choices B, C, and D are incorrect. Each of the expressions given in these answer choices gives W in terms of P and t but doesn't maintain the given relationship between W, P, and t. These expressions may result from performing different operations with t on each side of the equation. In choice B, W has been multiplied by t, and P has been divided by t. In choice C, W has been multiplied by t, and the quotient of P divided by t has been reciprocated. In choice D, W has been multiplied by t, and P has been added to t. However, in order to maintain the relationship between the variables in the given equation, the same operation must be performed with t on each side of the equation.